S2 Activation: Spectral Leakage and Equivariance in SO(3)-Equivariant Networks

A Numerical Study of Nonlinearity and Quadrature Effects

Experiments run on NERSC Perlmutter

1. Introduction

S2 Activation is the standard pointwise nonlinearity in SO(3)-equivariant neural networks, including SCN, eSCN, EquiformerV2, and related architectures. Given a set of spherical harmonic (SH) coefficients, it applies a scalar nonlinearity on the sphere and projects the result back to the SH basis. While the operation is exact in the limit of infinite bandwidth and infinite quadrature, practical implementations truncate the SH expansion at a maximum degree l_{max} and approximate the sphere integral with a finite quadrature rule. Both approximations introduce error and can break SO(3) equivariance.

This report investigates two orthogonal axes that control these errors:
(i) the choice of activation function (Section 2), and
(ii) the choice of quadrature / sampling strategy (Section 3).
We find that these two axes affect the system through fundamentally different mechanisms -- truncation error vs. aliasing error -- leading to a surprising dichotomy in their effects on coefficient accuracy vs. equivariance.

1.1 The S2 Activation Pipeline

Let $f : S^2 \to \mathbb{R}$ be a band-limited function represented by SH coefficients $\{c_{lm}\}$ for $l = 0, \ldots, l_{max}$ and $m = -l, \ldots, l$. The S2 Activation applies a pointwise nonlinearity σ in three steps:

Step 1 (Forward transform): Evaluate f on a set of N quadrature points $\{x_i\}$ on S^2:

$$f(x_i) = \sum_{l=0}^{l_{max}} \sum_{m=-l}^{l} c_{lm} Y_{lm}(x_i)$$

where Y_{lm} are real spherical harmonics.

Step 2 (Pointwise nonlinearity): Compute $g(x_i) = \sigma(f(x_i))$ at each point.

Step 3 (Inverse transform): Project back to SH coefficients via quadrature:

$$c'_{lm} = \sum_{i=1}^{N} w_i \, g(x_i) \, Y_{lm}(x_i)$$

where $\{w_i\}$ are the quadrature weights (summing to 4π).

1.2 Definitions

Angular power spectrum. For a function with SH coefficients $\{c_{lm}\}$, the power at degree l is:

$$P(l) = \sum_{m=-l}^{l} |c_{lm}|^2$$

This measures the total energy at angular frequency l.

Spectral leakage ratio. After applying σ to a band-limited input ($l \le l_{max}$), the output g generally has energy at all degrees. We define the leakage ratio as the fraction of output energy above the input bandwidth:

$$R = \sum_{l > l_{max}} P_g(l) \Big/ \sum_{l \ge 0} P_g(l)$$

where $P_g(l)$ is the power spectrum of g, computed using a high-resolution reference quadrature ($l_{max_ref} = 25$). R = 0 means no leakage; R = 1 means all energy leaked.

Coefficient error. Let c'_{out} be the output SH coefficients from S2 Activation with a given quadrature, and c'_{gt} the ground-truth coefficients obtained with a high-resolution reference quadrature (GL at $l_{max_ref} = 25$, truncated to l_{max}). The relative coefficient error is:

$$E_{coeff} = \|c'_{out} - c'_{gt}\| \Big/ \|c'_{gt}\|$$

Equivariance error. For a rotation R in SO(3) with Wigner-D matrix D(R), exact equivariance requires $\mathrm{S2Act}(D(R)\, c) = D(R)\, \mathrm{S2Act}(c)$. We measure the relative violation:

$$E_{equiv} = \|\mathrm{S2Act}(D(R)\, c) - D(R)\, \mathrm{S2Act}(c)\| \Big/ \|\mathrm{S2Act}(D(R)\, c)\|$$

averaged over 20 random rotations and 10 random inputs.

1.3 Experimental Setup

All experiments use random band-limited inputs: each SH coefficient c_{lm} is drawn i.i.d. from N(0,1), giving $(l_{max} + 1)^2$ total coefficients. Results are averaged over 10 random inputs (seeds 0-9). We test l_{max} in {3, 6, 10} and focus on $l_{max} = 6$ in the main

text.

Activation functions tested:
- C^inf smooth: Softplus(beta=1), SiLU, GELU, tanh
- C^0 (non-differentiable at origin): ReLU, abs, LeakyReLU
- Parametric family: Softplus(beta) with beta in {1, 3, 10, 30, 100}
  (Softplus(beta, x) = (1/beta) ln(1 + exp(beta x)); approaches ReLU as beta -> inf)
- Special: x^2 (exactly band-limited to 2 l_max), sin

Quadrature methods tested:
- Gauss-Legendre (GL): n_theta GL nodes in cos(theta) x n_phi uniform in phi.
  GL_kx denotes k-times oversampling (n_theta = k (l_max+1)).
- Lebedev: Symmetric quadrature exact for polynomials up to a given algebraic degree.
- Uniform (equiangular): Equally spaced grid in (theta, phi).

2. Equivariance Error vs. Activation Function

We first investigate how the choice of nonlinearity affects equivariance, holding the quadrature fixed. The key finding is that smoother activation functions produce less spectral leakage, and that leakage is strongly correlated with equivariance error.

2.1 Spectral Leakage Across Activations

Figure 1 shows the spectral leakage ratio R for each activation function at l_max = 3, 6, and 10. The leakage ratio measures the fraction of output energy that falls above the input bandwidth l_max (see Section 1.2). This energy is irrecoverably lost upon truncation back to l_max coefficients.

Activations are sorted from lowest to highest leakage. The pattern is consistent across all l_max values:
- C^inf activations (Softplus_1, SiLU, GELU) have the lowest leakage (R ~ 0.01-0.08), because their smoothness produces rapidly decaying high-l Fourier content.
- C^0 activations (ReLU, abs) have high leakage (R ~ 0.09-0.16), because the cusp at the origin generates power-law spectral tails.
- The Softplus(beta) family interpolates continuously between these extremes, confirming that the effect is controlled by activation smoothness.
- The polynomial x^2 has zero leakage beyond 2 l_max by construction (its output is exactly band-limited).
- tanh shows moderate leakage despite being C^inf. This is because tanh saturates to +/-1, compressing the output range and concentrating energy differently from non-saturating smooth activations.

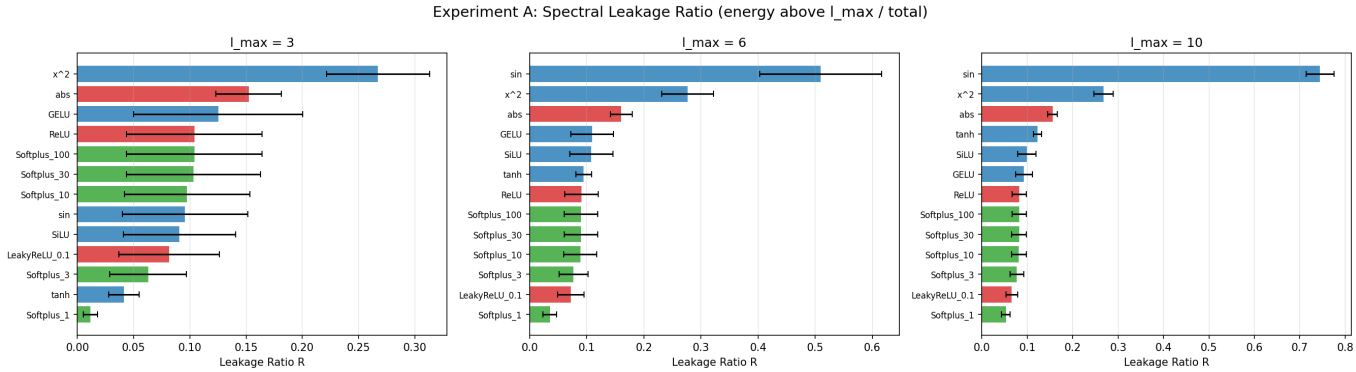

Figure 1: Spectral leakage ratio R = sum_{l>l_max} P(l) / sum_l P(l) for each activation function, measured at l_max = 3, 6, 10. Input: 10 random SH coefficient vectors (c_lm ~ N(0,1)). Output spectrum computed via GL quadrature at l_max_ref = 25. Red bars: C^0 activations. Green: Softplus family. Blue: C^inf smooth. Error bars: standard deviation across 10 inputs.

2.2 Leakage Predicts Equivariance Error

Figure 2 plots the equivariance error E_equiv (measured in Experiment C) against the spectral leakage ratio R (measured in Experiment A) for each activation at l_max = 6, using GL_1x quadrature.

The strong positive correlation confirms the mechanistic link: activations that leak more energy beyond l_max suffer larger equivariance violations, because the leaked energy aliases back into the l <= l_max coefficients in a rotation-dependent manner. Since different rotations move the quadrature points to different positions on the sphere, the aliasing pattern changes with rotation, breaking equivariance.

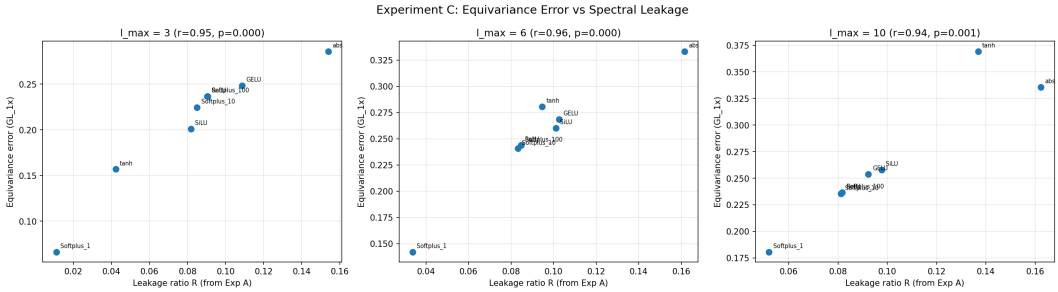

Figure 2: Equivariance error E_equiv vs. spectral leakage ratio R for each activation at l_max = 3, 6, 10 using GL_1x quadrature. E_equiv = ||S2Act(Dc) - D S2Act(c)|| / ||S2Act(Dc)||, averaged over 20 random SO(3) rotations and 10 random inputs. Each point is one activation. The correlation shows that spectral leakage is a strong predictor of equivariance violation.

Table 1: Leakage and Equivariance Error (l_max = 6, GL_1x)

Activation	Leakage R	Equiv Error	Smoothness
Softplus_1	0.0354	0.1419	C^inf
SiLU	0.1079	0.2600	C^inf
GELU	0.1096	0.2684	C^inf
Softplus_10	0.0887	0.2407	C^inf
ReLU	0.0903	0.2437	C^0
Softplus_100	0.0903	0.2437	C^inf
tanh	0.0944	0.2807	C^inf (sat.)
abs	0.1604	0.3333	C^0

3. Equivariance Error vs. Sampling Strategy

We now hold the activation function fixed and vary the quadrature. A surprising dichotomy emerges: oversampling has almost no effect on coefficient accuracy, but dramatically reduces equivariance error. We also compare different quadrature families (Gauss-Legendre, Lebedev, and uniform grids).

3.1 The Truncation-Aliasing Dichotomy

Figure 3 shows the relative coefficient error E_coeff as a function of GL oversampling ratio (1x, 2x, 3x) for each activation. The error is nearly flat: increasing the number of quadrature points from ~100 to ~700 barely changes the output coefficients.

This is because coefficient error is dominated by truncation, not aliasing. When the nonlinearity generates energy at degrees l > l_max, that energy is lost regardless of how accurately we integrate -- the output SH basis simply cannot represent it. More quadrature points give a more accurate integral, but the integral is still projecting onto a truncated basis, so the irrecoverable truncation error dominates.

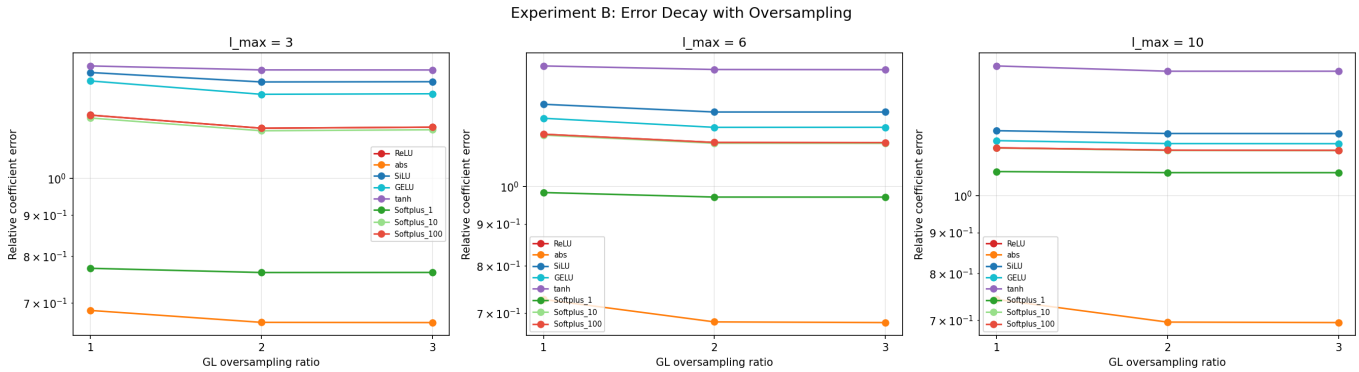

Figure 3: Relative coefficient error E_coeff = ||c_out - c_gt|| / ||c_gt|| vs. GL oversampling ratio (1x, 2x, 3x) for each activation at l_max = 3, 6, 10. Ground truth: GL quadrature at l_max_ref = 25, truncated to l_max. The near-flat curves show that increasing quadrature resolution does not reduce coefficient error, because the error is dominated by truncation of high-l content, not by quadrature aliasing.

In contrast, Figure 4 shows the equivariance error E_equiv as a function of GL oversampling ratio. Here, oversampling produces dramatic improvements: GL_3x reduces equivariance error by 10x to 450x compared to GL_1x, depending on the activation.

Why does oversampling help equivariance but not coefficient accuracy? The key insight is that truncation error is equivariant but aliasing error is not:

- Truncation is a projection: it discards all l > l_max components. This projection commutes with rotation (since each degree l transforms independently under SO(3)), so truncation alone does not break equivariance. It loses information, but it loses the same information regardless of orientation.

- Aliasing is quadrature-dependent: when the quadrature rule cannot exactly integrate products of high-l and low-l harmonics, high-frequency energy folds back into low-l coefficients. The folding pattern depends on the specific positions of quadrature points relative to the function being integrated. Since rotating the input moves the function relative to the fixed quadrature grid, the aliasing pattern changes with rotation, breaking equivariance.

Oversampling reduces aliasing by providing enough quadrature points to accurately integrate the cross-terms between high-l and low-l harmonics. This is why smooth activations benefit more from oversampling: they have less high-l content to alias in the first place, so moderate oversampling can effectively eliminate aliasing for them.

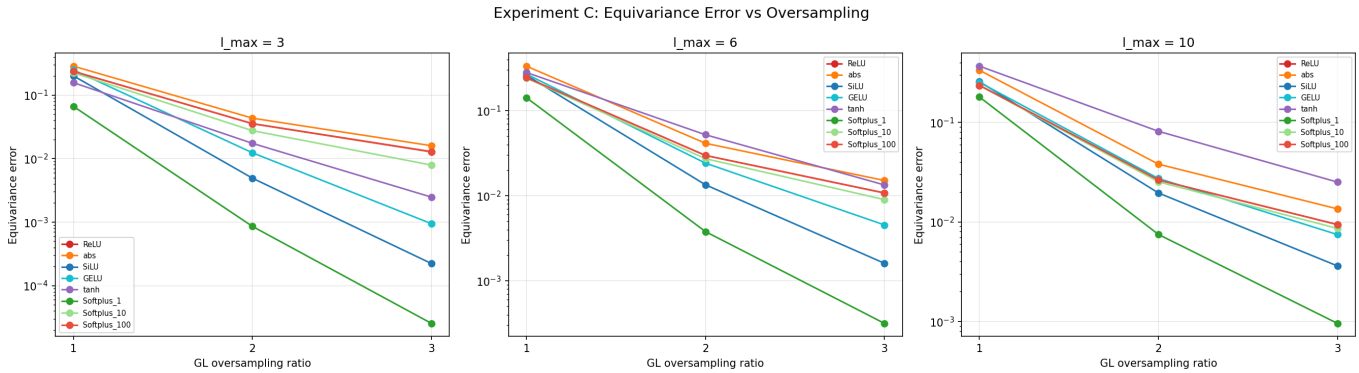

Figure 4: Equivariance error E_equiv vs. GL oversampling ratio (1x, 2x, 3x) for each activation at l_max = 3, 6, 10. Compare with Figure 3: while coefficient error is insensitive to oversampling, equivariance error drops by 1-2 orders of magnitude. This is because aliasing (not truncation) is the dominant source of equivariance violation, and oversampling directly reduces aliasing. Smooth activations (Softplus_1, SiLU) benefit most.

Table 2: Equivariance Error vs. GL Oversampling (l_max = 6)

Activation	GL_1x (N~98)	GL_2x (N~338)	GL_3x (N~722)	Improvement
Softplus_1	1.42e-01	3.78e-03	3.15e-04	450x
SiLU	2.60e-01	1.34e-02	1.60e-03	162x
GELU	2.68e-01	2.41e-02	4.53e-03	59x
Softplus_10	2.41e-01	2.72e-02	8.98e-03	27x
ReLU	2.44e-01	2.97e-02	1.08e-02	23x
Softplus_100	2.44e-01	2.97e-02	1.07e-02	23x
tanh	2.81e-01	5.20e-02	1.34e-02	21x
abs	3.33e-01	4.13e-02	1.51e-02	22x

3.2 Comparison of Quadrature Methods

Beyond oversampling within a single quadrature family, we compare three qualitatively different quadrature methods: Gauss-Legendre (GL), Lebedev, and uniform (equiangular) grids. Figure 5 shows the equivariance error for all six sampling configurations tested at l_max = 6.

Key observations:

- At matched oversampling (GL_1x vs. Leb_min), GL slightly outperforms Lebedev, likely because GL nodes are optimally placed for the tensor-product structure of S^2.
- Lebedev at 2x (N = 230) performs comparably to GL_2x (N = 338) for smooth activations, achieving similar equivariance error with fewer points.
- Uniform grids at high resolution (N = 5000) achieve the lowest equivariance error across all activations, but at much higher cost. This confirms that the error reduction is a genuine quadrature effect, not an artifact of GL-specific structure.
- The relative ranking of activations is preserved across all quadrature methods: smoother activations consistently achieve lower equivariance error.

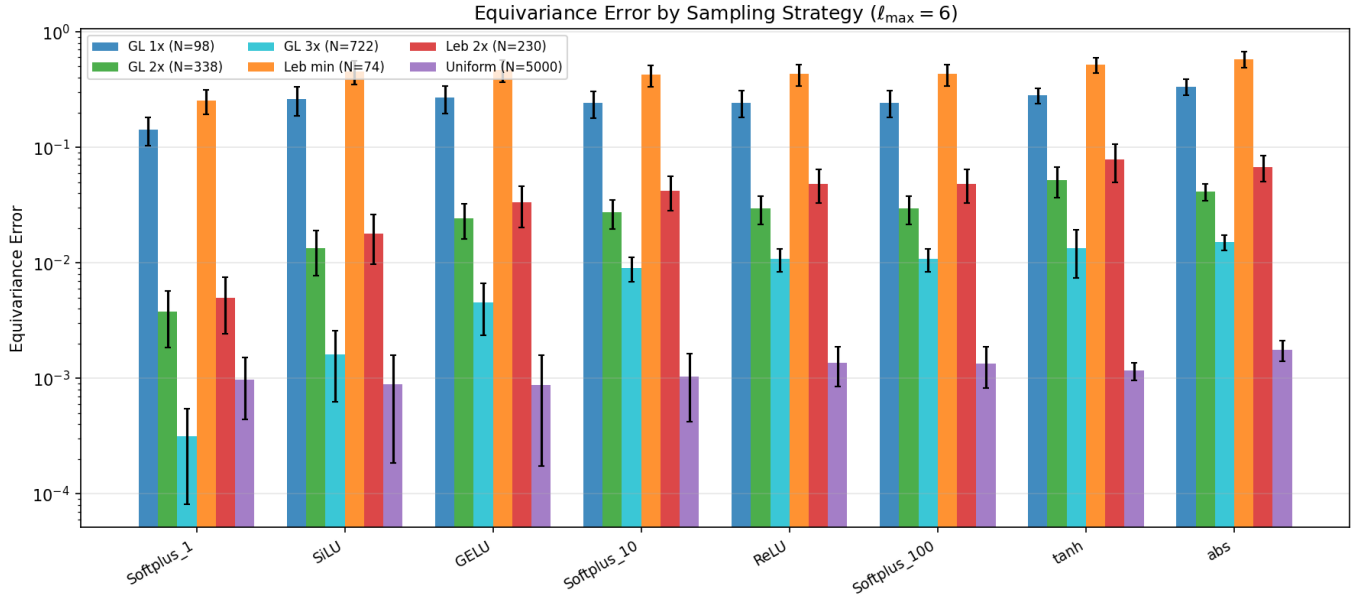

Figure 5: Equivariance error E_equiv for each activation under six sampling strategies at l_max = 6. N denotes the number of quadrature points. GL_kx: Gauss-Legendre with k-times oversampling. Leb min/2x: Lebedev quadrature at minimum / double degree. Uniform: equiangular grid. Log scale. Error bars: std over 20 rotations x 10 inputs. The activation ranking is stable across methods, confirming that smoothness effects are independent of quadrature choice.

Table 3: Equivariance Error Across Sampling Methods (l_max = 6)

Activation	GL_1x	GL_2x	Leb_min	Leb_2x	Uniform
N (points)	98	338	74	230	5000
Softplus_1	1.42e-01	3.78e-03	2.54e-01	4.98e-03	9.80e-04
SiLU	2.60e-01	1.34e-02	4.53e-01	1.79e-02	8.86e-04
ReLU	2.44e-01	2.97e-02	4.29e-01	4.85e-02	1.36e-03
tanh	2.81e-01	5.20e-02	5.17e-01	7.80e-02	1.17e-03
abs	3.33e-01	4.13e-02	5.79e-01	6.75e-02	1.76e-03

4. Preliminary: Downstream Task Performance

Goal: Explore whether spectral/equivariance analysis correlates with downstream performance.

Method: Train a 1-layer SphericalCNN on a synthetic 5-class classification task (l_max = 6), where class-discriminative information resides in high-l coefficients. The model consists of a linear layer, S2Activation, and a classification head. We test 5 activations x 3 sampling configs, with 2 runs per configuration (15 epochs each).

Result (PRELIMINARY): Accuracy differences across activations are small (1-2%). Notably, tanh achieves the best task accuracy despite having worse leakage and equivariance than Softplus_1 or SiLU. This suggests that activation expressivity and optimization landscape effects are significant confounders that this experiment does not control for. The smoothness-to-performance chain is not validated by these results alone.

Specifically, different nonlinearities have different approximation capacity: tanh can represent a richer class of functions on S^2 than, say, a very smooth Softplus(1), even when both suffer from truncation. A controlled experiment would need to match expressivity across activations (e.g., by adjusting model width or depth) before attributing accuracy differences to equivariance alone.

Table 4: Test Accuracy

Activation	GL_1x	GL_2x	Leb_min
tanh	0.875	0.873	0.868
Softplus_1	0.873	0.870	0.871
SiLU	0.863	0.863	0.865
ReLU	0.863	0.863	0.863
Softplus_10	0.860	0.864	0.865

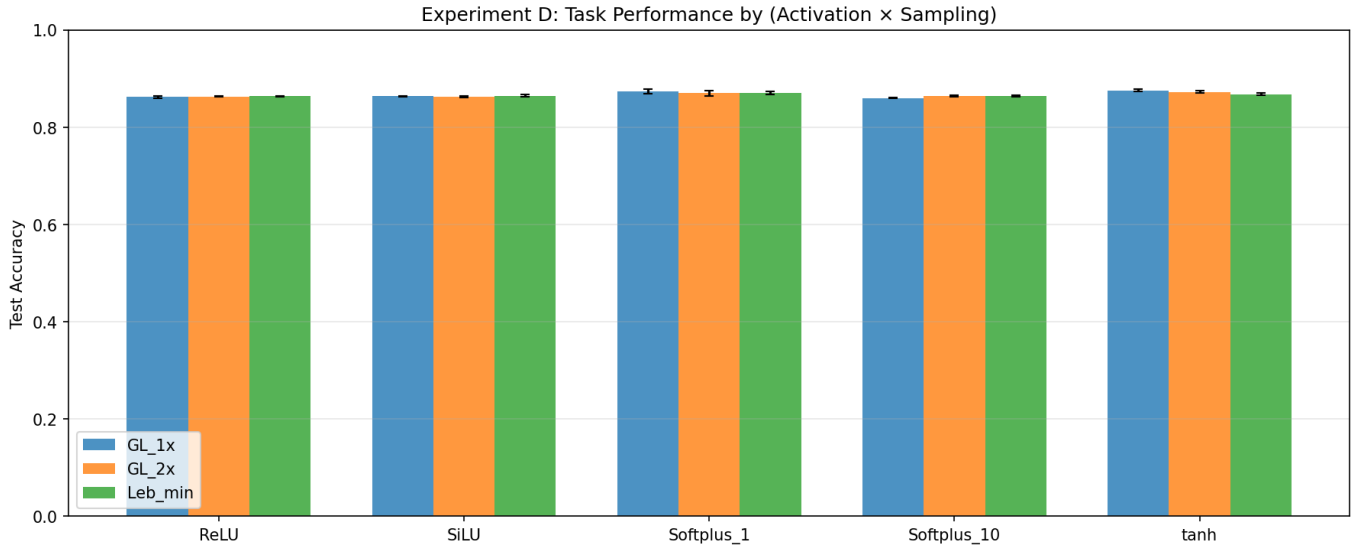

Figure 6: Test accuracy by (activation x sampling) configuration. 2 runs per config, 15 epochs, l_max = 6. Error bars: std across runs.

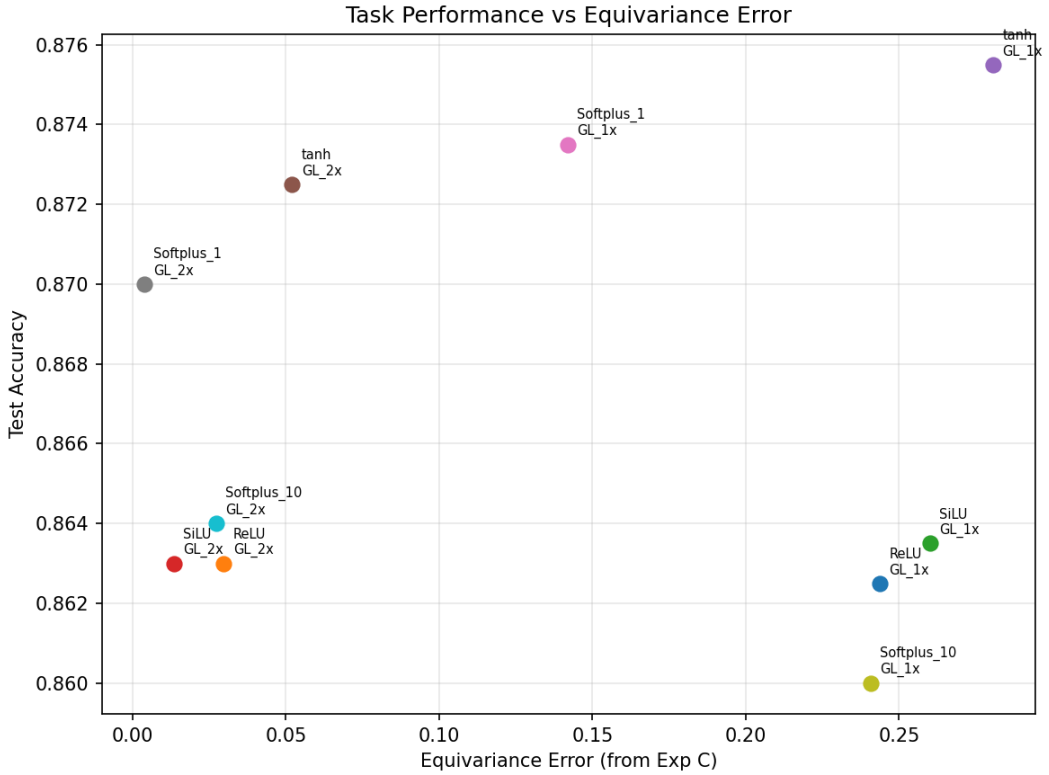

Figure 7: Task accuracy vs. equivariance error. If the smoothness -> equivariance -> performance chain held, we would expect a negative correlation. The scatter shows no clear monotonic trend, suggesting that expressivity confounds the relationship.

5. Conclusions

This study identifies two independent axes controlling the behavior of S2 Activation and reveals a fundamental dichotomy in their error mechanisms.

1. ACTIVATION SMOOTHNESS CONTROLS EQUIVARIANCE VIA SPECTRAL LEAKAGE
Smoother activations (Softplus_1, SiLU) produce less spectral leakage, and leakage is strongly correlated with equivariance error (Figure 2). The Softplus(beta) family provides a continuous 1-parameter knob from smooth (beta = 1, low leakage) to sharp (beta -> inf ~ ReLU, high leakage). This relationship is consistent across all l_max values and sampling methods.

2. THE TRUNCATION-ALIASING DICHOTOMY
Coefficient error is dominated by truncation (insensitive to oversampling, Figure 3), while equivariance error is dominated by aliasing (highly sensitive to oversampling, Figure 4). This is because truncation commutes with rotation (it projects out the same degrees regardless of orientation), but aliasing does not (the fold-back pattern depends on the function's orientation relative to the quadrature grid).

3. QUADRATURE METHOD MATTERS LESS THAN OVERSAMPLING
GL, Lebedev, and uniform grids produce similar equivariance error at matched point counts (Figure 5). The dominant factor is how many points are used, not their specific arrangement.

4. TASK PERFORMANCE REQUIRES FURTHER INVESTIGATION
Our preliminary Experiment D does not validate a smoothness-to-performance chain. Activation expressivity is a likely confounder: different nonlinearities have different approximation capacity on S^2, which can outweigh equivariance effects in a shallow model. Controlled experiments matching expressivity across activations are needed before drawing conclusions about downstream impact.

5. PRACTICAL GUIDANCE (for equivariance)
- To minimize equivariance error: use Softplus(1) or SiLU + >= 2x GL oversampling.
- Cost-effective default: SiLU + GL_1x (standard in most models).
- Parametric control: Softplus(beta) to trade smoothness vs. sharpness.
- Caveat: minimizing equivariance error does not guarantee best task performance.

Appendix A: Additional Spectral Leakage Figures

These figures provide additional detail on the spectral structure of different activations beyond the summary in Figure 1.

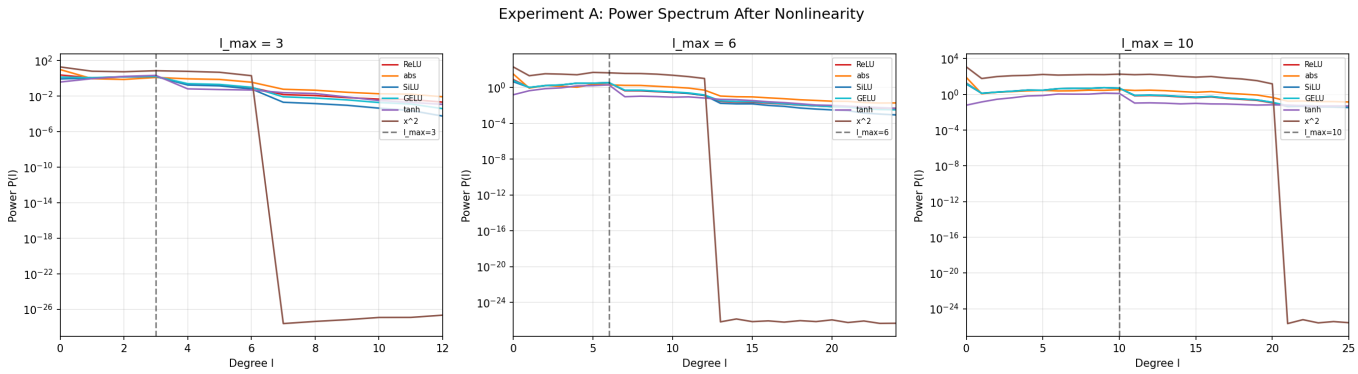

Figure A1: Full power spectrum P(l) vs. degree l after applying each nonlinearity to a random band-limited input (l <= l_max). Dashed vertical line marks l_max. Energy to the right of this line is lost upon truncation. Panels: l_max = 3, 6, 10. Log scale.

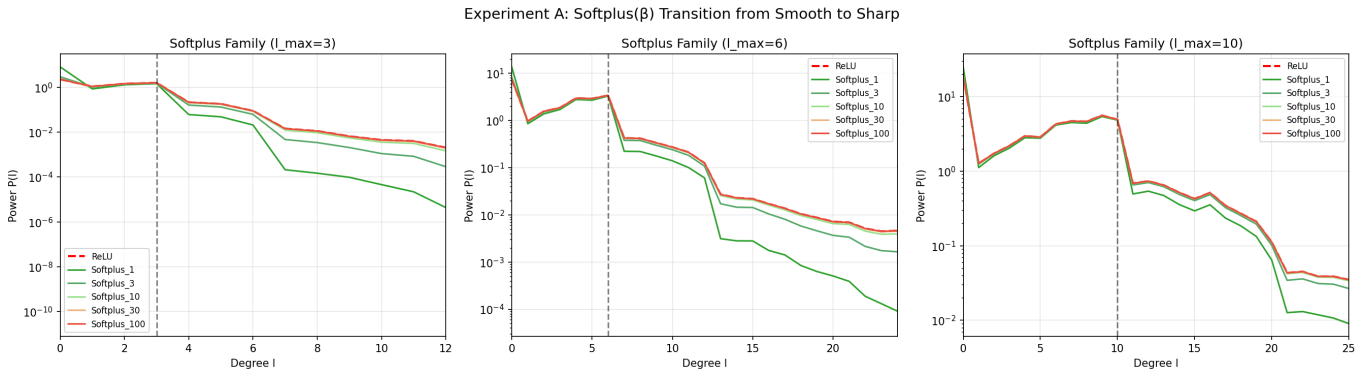

Figure A2: Power spectra for the Softplus(beta) family with ReLU reference (dashed red). As beta increases, the spectrum approaches the ReLU power-law tail. Panels: l_max = 3, 6, 10.

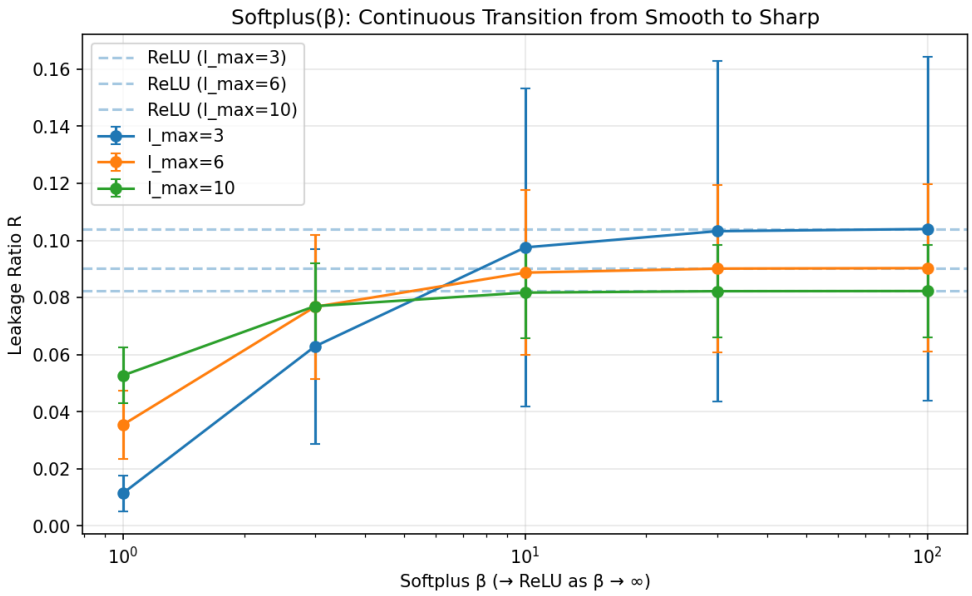

Figure A3: Leakage ratio R vs. Softplus beta on a log scale, for l_max = 3, 6, 10. Horizontal dashed lines show the corresponding ReLU leakage. Confirms continuous interpolation from smooth to sharp.

Table A1: Leakage Ratio R (Full)

Activation	R (l_max=3)	R (l_max=6)	R (l_max=10)

Softplus_1	0.0114	0.0354	0.0527
Softplus_3	0.0629	0.0767	0.0770
SiLU	0.0906	0.1079	0.0990
GELU	0.1252	0.1096	0.0929
LeakyReLU_0.1	0.0816	0.0721	0.0658
ReLU	0.1041	0.0903	0.0823
Softplus_10	0.0976	0.0887	0.0817
Softplus_30	0.1032	0.0901	0.0822
Softplus_100	0.1040	0.0903	0.0823
tanh	0.0413	0.0944	0.1227
sin	0.0958	0.5095	0.7447
abs	0.1522	0.1604	0.1558
x^2	0.2673	0.2769	0.2676

Appendix B: Additional Coefficient Error Figures

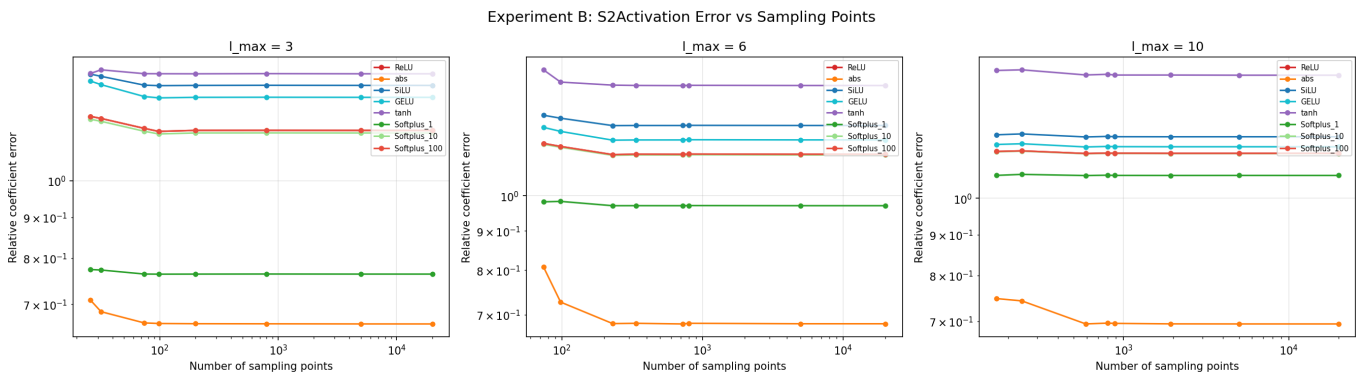

Figure B1: Coefficient error vs. number of sampling points for each activation. Even at N = 20000, error plateaus at the truncation floor. Only smoother activations achieve lower error.

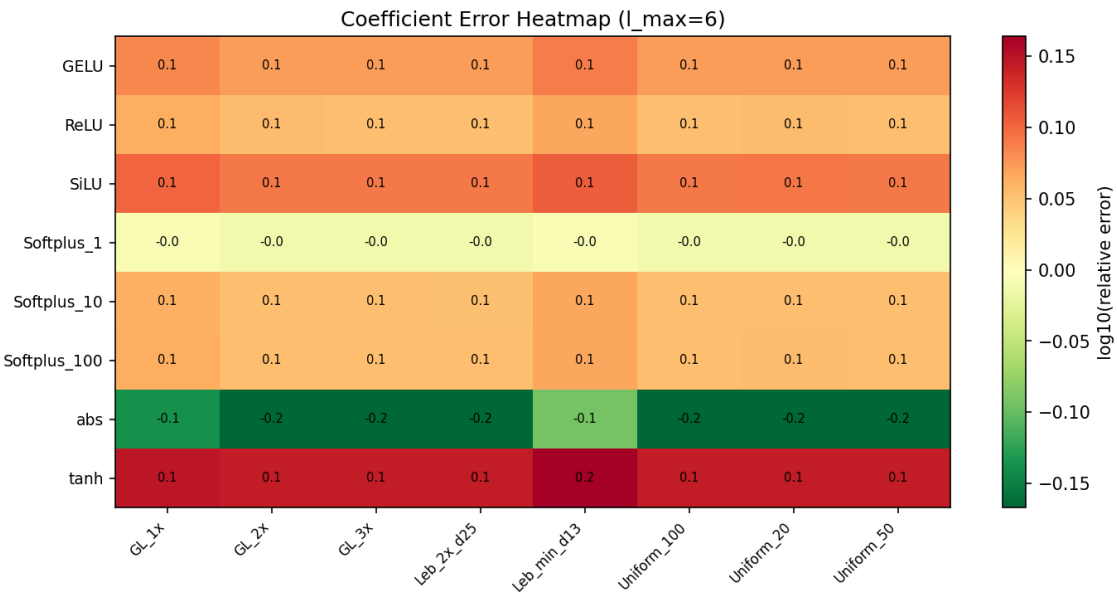

Figure B2: Coefficient error heatmap (activation x sampling method) at l_max = 6. Rows: activations. Columns: sampling configurations. Color: relative error. Confirms that the activation axis drives error, not sampling.

Table B1: Coefficient Error (l_max = 6)

Activation	GL_1x	GL_2x	GL_3x	Trunc Ratio
Softplus_1	0.983	0.971	0.971	0.1853
SiLU	1.262	1.234	1.234	0.3232
GELU	1.213	1.182	1.182	0.3264
ReLU	1.159	1.133	1.132	0.2966
Softplus_10	1.156	1.130	1.130	0.2939
tanh	1.406	1.392	1.391	0.3065
abs	0.727	0.682	0.681	0.3998
Softplus_100	1.159	1.133	1.132	0.2966

Appendix C: Additional Equivariance Figures

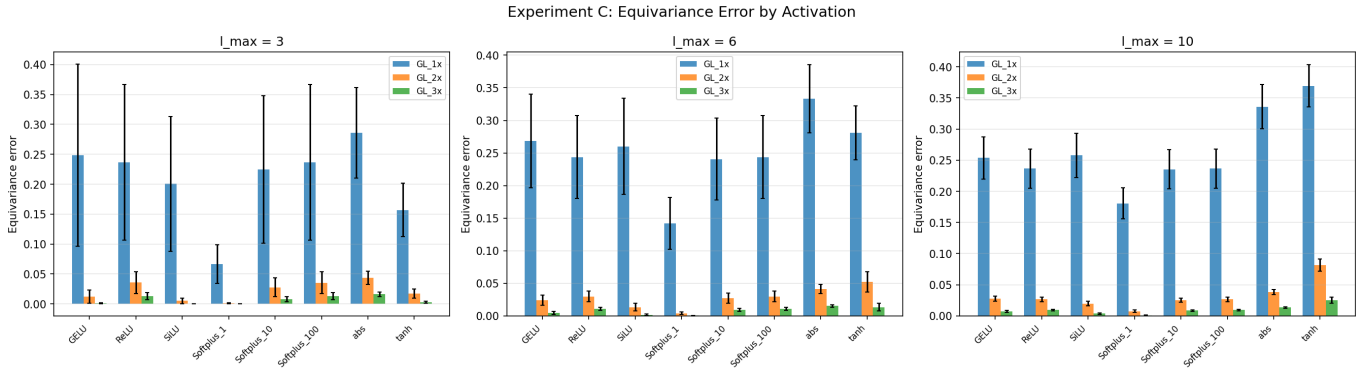

Figure C1: Equivariance error grouped by activation, with separate bars for GL_1x, GL_2x, GL_3x. Panels: l_max = 3, 6, 10.

Appendix D: Phase 1 - Quadrature Method Comparison

Phase 1 compared quadrature methods (GL, Lebedev, Uniform, Fibonacci) for pure SH reconstruction accuracy without nonlinearities.

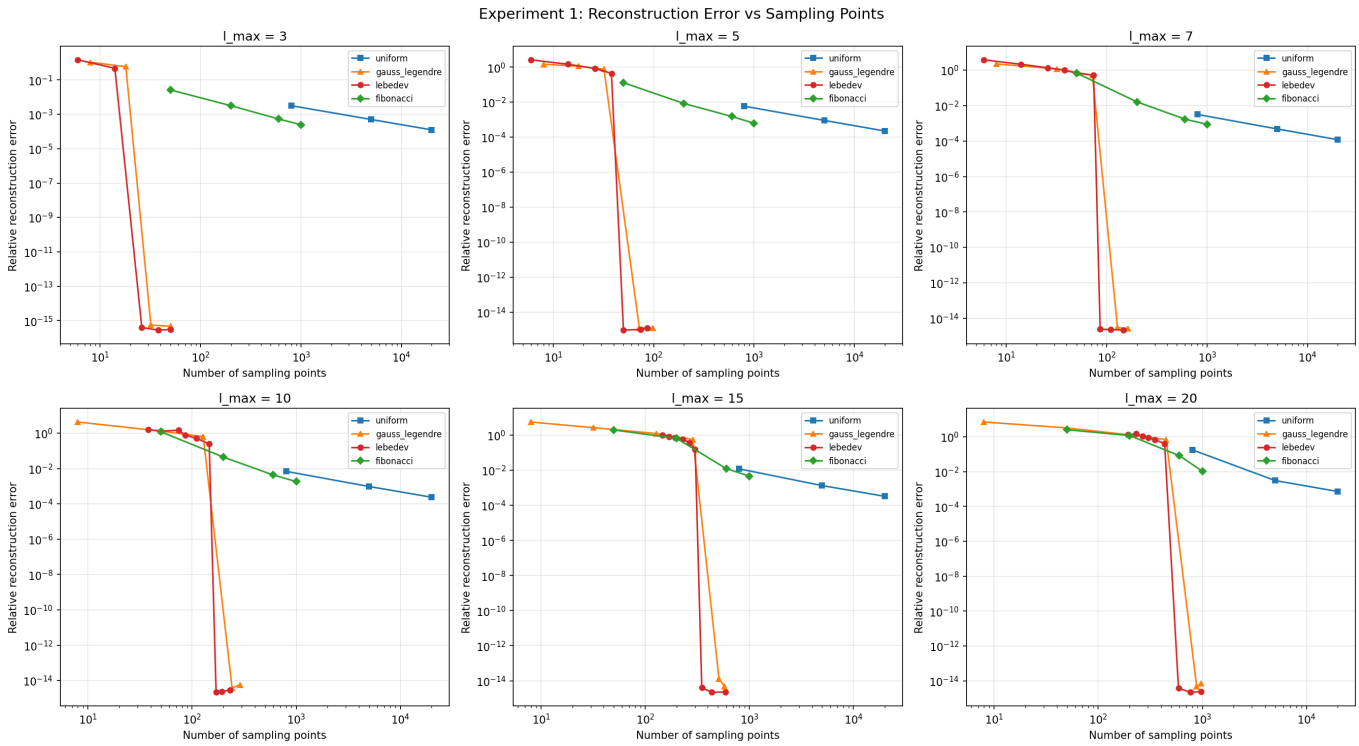

Figure D1: SH reconstruction error vs sampling points

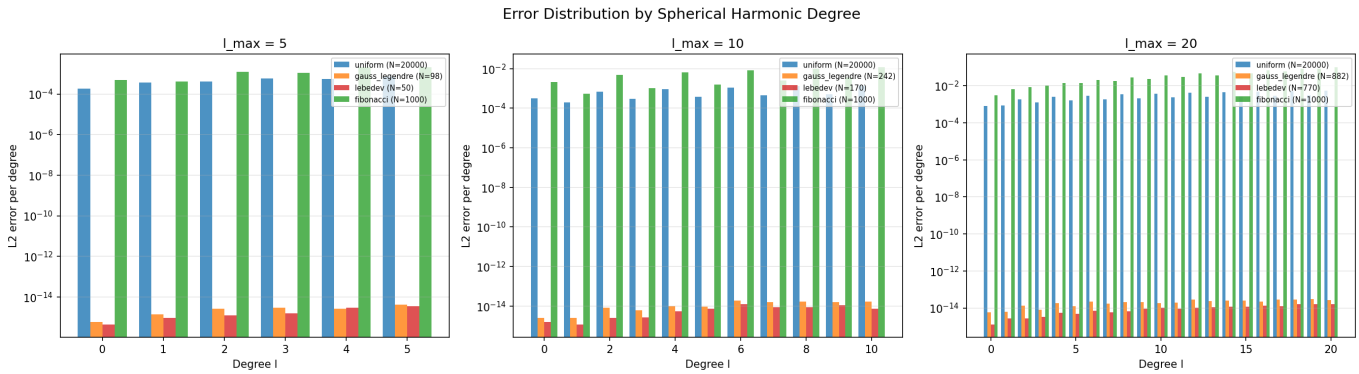

Figure D2: Per-degree reconstruction error

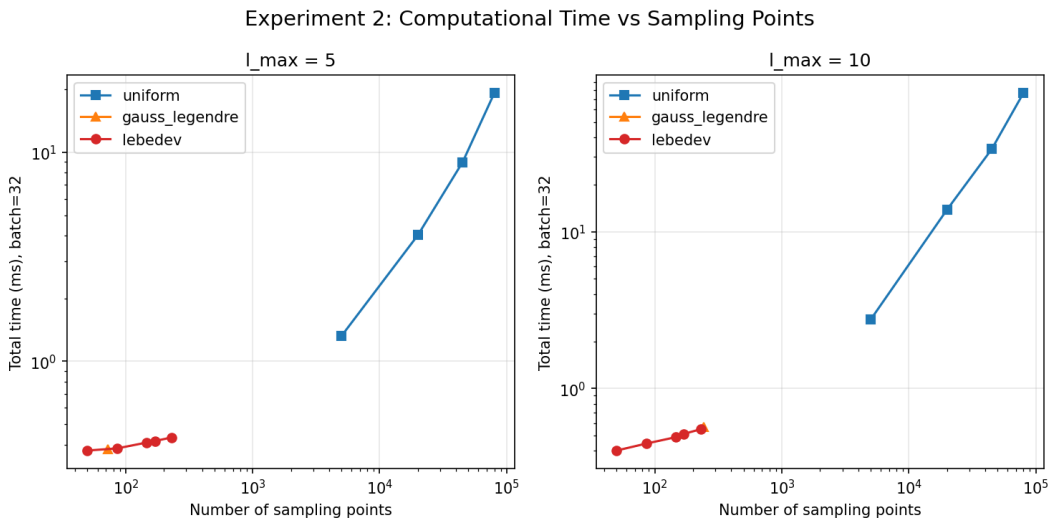

Figure D3: Computational cost scaling

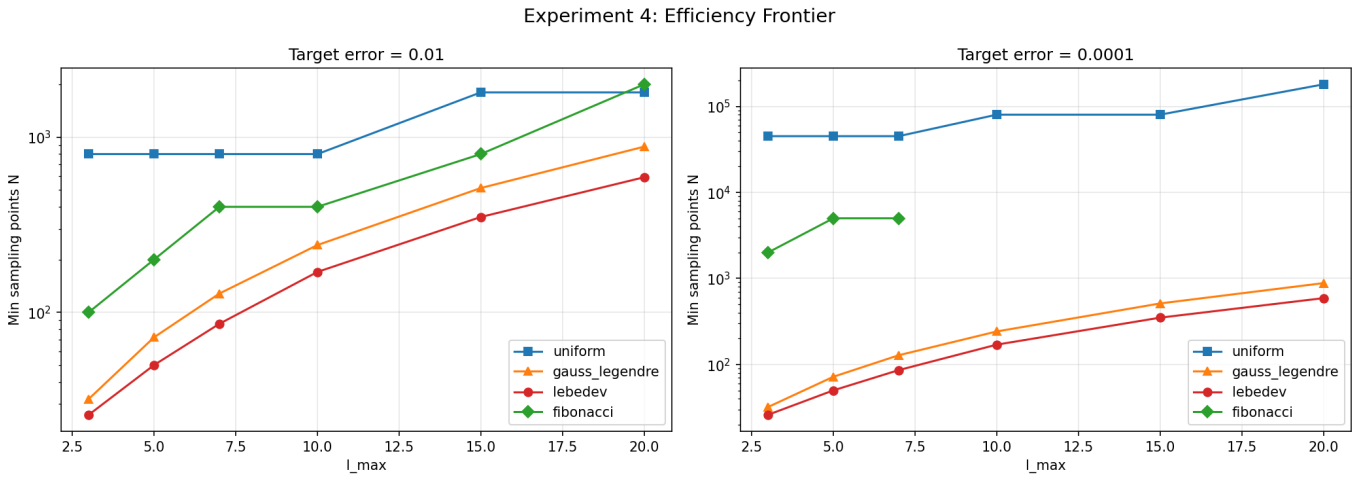

Figure D4: Efficiency frontier

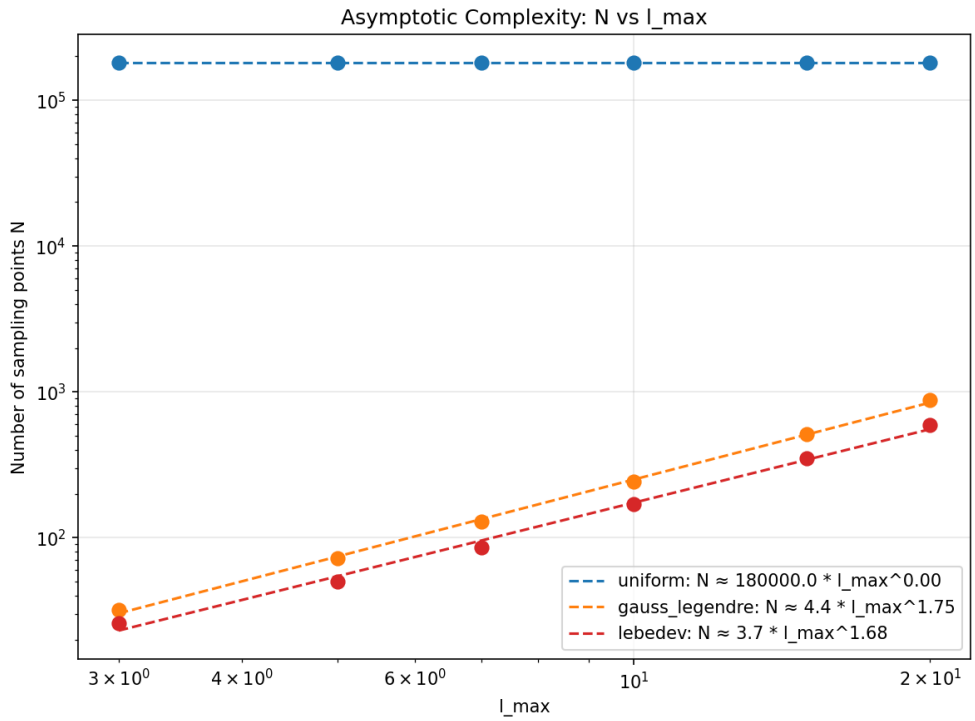

Figure D5: Asymptotic complexity